Slide 1: SIGNALFIRE AI

Find the warning signs hidden across your data

- **Category:** AI-Powered Business Intelligence, Anomaly Detection & Cross-Dataset Risk Discovery
- **Domain:** Enterprise Retail Operations & Supply Chain Telemetry
- **Built with:** Exasol Personal, DuckDB Fallback, FastAPI, React, and Robust Multi-Metric Analytics
- **Mission:** Help operations analysts discover unusual cross-dataset patterns and prioritize emerging risks before they escalate.

Slide 2: The Operational Problem

Why siloed operational metrics obscure emerging systemic failures

- **Siloed Monitoring:** Sales, deliveries, complaints, refunds, and retention are analyzed separately.
- **Developing Crises:** Small, sub-threshold changes across individual metrics co-occur to form severe failures.
- **High Review Burden:** Manual review is time-consuming and heavily dependent on uncoordinated single-metric alerts.
- **Target Audience:** Business operations managers, supply chain analysts, and performance leaders.

Slide 3: Why Existing Dashboards Miss Cross-Metric Risks

Independent thresholds fail when small changes co-occur across tables

- **Independent Alerting:** Delivery alerts fire only on large transit delays, missing early warning signals in complaints.
- **Temporal Lag Disconnection:** Transit delays lead return requests by 3-4 days; unaligned dashboards miss the lead-lag offset.

- **Seasonality Noise:** Static percentage thresholds generate false alarms during normal weekend volume surges.
- **No Control Cohorts:** Lacks automated contrast against unaffected operational segments (e.g. North vs South/East/West).

Slide 4: The SIGNALFIRE AI Solution

A principled 5-stage analytical pipeline separating facts from hypotheses

- **1. Independent Anomaly Detection:** Robust Z-Scores via Median Absolute Deviation (MAD) and IQR fences.
- **2. Multi-Dataset Temporal Alignment:** Synchronizing 8 tables onto unified 24h temporal windows.
- **3. Statistical Association Engine:** Empirical Pearson and Spearman correlation with lead-lag offset estimation.
- **4. Unaffected Cohort Benchmarking:** Automated contrast against non-incident regional controls.
- **5. Disciplined Synthesis:** Explicit separation of Facts, Statistical Associations, Hypotheses, and Limitations.

Slide 5: Product Walkthrough & Demonstration

A unified decision-support workspace for analysts and leaders

- **Overview Dashboard:** Real-time KPI cards, weekly signal trend curves, risk breakdown, and active engine badge.
- **Data Explorer & Profiler:** Schema inspector, null distributions, paginated record preview, and safe CSV upload.
- **Signal Detection Engine:** Complete evidence trails, observed vs baseline values, and detection methodologies.
- **Cross-Metric Engine:** Prioritized risk clusters, empirical correlation metrics, and control cohort benchmarks.
- **Interactive Risk Map & Workspace:** Topological segment graph, priority assignments (P1-P3), and conclusion logging.

- **What-If Stress Testing & Export:** Arithmetic vs model projections, and one-click Executive PDF/JSON/CSV downloads.

Slide 6: Technical Architecture & Exasol Integration

High-throughput in-memory SQL analytics with transparent local fallback

- **Analytical Storage:** Exasol Personal integration via official pyexasol driver (WebSocket JSON-RPC).
- **Adapter Architecture:** Abstract DatabaseAdapter supporting Exasol Personal and DuckDB OLAP fallback.
- **Full Transparency:** Live engine status displayed across UI and API. Never mislabels local calculations.
- **Backend Engine:** Python 3.12, FastAPI, Pydantic v2, Pandas, NumPy, Scikit-learn, SciPy, and ReportLab.
- **Frontend Experience:** React 18, TypeScript, Vite, Tailwind CSS, Recharts, and 3D glassmorphism design language.

Slide 7: Innovation & Differentiating Principles

Credible, research-backed capabilities without hype or false claims

- **Correlation != Causation:** System never claims correlation proves causation. Explicitly formulates competing hypotheses.
- **Robust MAD Normalization:** Median Absolute Deviation suppresses extreme noise that distorts standard deviations.
- **Automated Control Cohort Validation:** Verifies localized shocks against unaffected regional cohorts.
- **Evidence-Backed AI Assistant:** Tool-grounded query architecture that executes verified SQL analytics.

Slide 8: Analytical Evaluation & Empirical Metrics

Evaluated strictly against recorded ground-truth synthetic anomaly injections

- **SignalFire Multi-Metric Detector:** Precision = 71.4% | Recall = 100.0% | F1-Score = 0.833 (Zero false negatives).
- **Simple %-Change Baseline Detector:** Precision = 45.5% | Recall = 100.0% | F1-Score = 0.625 (High false positive rate).
- **Detection Latency:** Signals isolated within 1 to 3 observation windows of onset.
- **Honest Limitations:** Ground truth established on synthetic data; non-stationary production distributions require ongoing calibration.

Slide 9: Target Users & Operational Impact

Empowering cross-functional teams to act before operational issues compound

- **Retail Operations Managers:** Detect carrier bottlenecks before transit delays trigger customer return waves.
- **Supply Chain Analysts:** Connect warehouse stockouts directly to upstream supplier defect and transit delays.
- **Customer Experience Teams:** Mitigate repeat purchase churn by identifying fulfillment root causes.
- **Leadership & Executives:** Receive audit-ready PDF reports with verified conclusions for SLA partner negotiations.

Slide 10: Future Roadmap & Closing

Scaling cross-dataset intelligence for enterprise operations

- **Distributed Exasol Clusters:** Scaling from Exasol Personal to multi-node clusters for billion-record datasets.
- **Multimodal Telemetry Fusion:** Incorporating unstructured customer support tickets and dispatch logs.

- **Automated ERP Orchestration:** Triggering automated supplier notifications and SLA penalty workflows in SAP/NetSuite.
- **SIGNALFIRE AI:** Find the warning signs hidden across your data.